

Project Specifications – Group 21

Project Topic - AI-Powered Credit Default Risk Prediction System Using Machine Learning and Neural Network Techniques

Team Github Link: <https://github.com/REKNOS/AI-Powered-Credit-Default-Risk-Prediction-System>

1. Project Topic:

This project aims to develop an AI-Powered Credit Default Risk Prediction System using Machine Learning and Neural Network Techniques to identify customers who are likely to default on credit obligations. The proposed system will analyse customer financial behaviour, payment history, and demographic information to support informed lending decisions, reduce financial losses, and improve credit risk management. Multiple machine learning and neural network models will be implemented and compared to identify the most effective approach for credit default prediction, while Explainable AI techniques will be used to enhance model transparency and interpretability.

2. Research Question / Problem Statement:

Credit default remains a significant challenge for financial institutions worldwide. Inaccurate credit assessments can result in substantial financial losses, increased operational risk, and reduced customer trust. Traditional rule-based approaches often struggle to identify hidden patterns within large-scale financial datasets. Although machine learning has improved predictive performance, many models lack transparency and interpretability. Therefore, there is a need for an intelligent and explainable system capable of accurately predicting customer credit default risk while providing meaningful insights into the factors influencing predictions.

3. Objectives:

- Analyse customer financial behaviour using structured credit datasets.
- Perform preprocessing, cleaning, and feature engineering.
- Conduct exploratory data analysis and visualisation.
- Compare multiple machine learning and neural network algorithms.
- Evaluate performance using Accuracy, Precision, Recall, F1-Score and ROC-AUC.
- Apply Explainable AI techniques using SHAP and LIME.
- Develop a reproducible and well-documented workflow.

4. Goal / Target:

The primary goal is to develop a reliable intelligent system capable of predicting potential customer credit defaults with high accuracy. The proposed solution aims to reduce financial losses, improve lending decisions, strengthen credit risk assessment strategies, and provide transparent AI-driven insights for financial institutions. The project also focuses on comparing multiple Machine Learning, Deep Learning, Hybrid, and Transfer Learning models to identify the most effective approach for credit default prediction. Furthermore, the project aims to develop a user-friendly web application that enables real-time credit risk analysis and supports automated, data-driven financial decision-making in real-world banking and financial environments.

5. Key Performance Indicators (KPI):

KPI / Metric	Target Value	Description
Accuracy	≥ 85%	Measures the overall correctness of model predictions
Precision	≥ 80%	Ensures predicted defaulters are correctly identified
Recall	≥ 80%	Measures the ability to detect high-risk customers
F1-Score	≥ 80%	Balances Precision and Recall for reliable performance
ROC-AUC Score	≥ 0.85	Evaluates the overall classification capability of the model
False Positive Rate	≤ 15%	Minimises incorrect classification of safe customers as risky
False Negative Rate	≤ 15%	Reduces the risk of missing actual default customers
Explainable AI Coverage	100% SHAP/LIME Support	Ensures model predictions are interpretable and transparent
Model Comparison	Minimum 6 ML/DL Models	Enables performance benchmarking across multiple models
Data Quality	100% Missing Values Addressed	Ensures datasets are clean and suitable for training

6. Datasets and Resource Links:

Dataset 1 – [Default of Credit Card Clients Dataset](#)

This dataset contains financial information, payment history, demographic attributes, bill statements, repayment records, and default payment status for approximately 30,000 credit card customers. It is widely used in credit risk prediction research.

Dataset 2 – [Statlog German Credit Dataset](#)

This dataset contains customer financial profiles and classifies individuals into good and bad credit risk categories. It is extensively used for benchmarking classification algorithms and validating predictive credit risk models.

7. Ethical Considerations:

- The datasets used in this project are publicly available academic datasets obtained from the UCI Machine Learning Repository.
- All datasets are anonymised and do not contain personally identifiable information (PII).
- The project does not involve web scraping, social media data collection, or unauthorised access to private user information.
- The project complies with GDPR principles as no sensitive personal data is processed or stored.
- The project follows University ethical policies related to responsible AI development and data usage.
- Bias detection and fairness evaluation techniques will be applied to minimise discriminatory predictions in credit risk assessment.
- All project data, models, and source code will be securely stored and managed using GitHub version control and access management practices.
- The developed AI system is intended only as a decision-support tool and will not replace human judgement in financial lending decisions.

8. Tools and Technologies:

Technology / Tool	Description
Python	Primary programming language used for AI and data science development
Jupyter Notebook	Interactive environment for coding, experimentation, and model analysis
Pandas	Used for data manipulation and preprocessing
NumPy	Supports numerical computations and matrix operations
Matplotlib	Used for data visualisation and graphical analysis
Seaborn	Provides advanced statistical data visualisations
Scikit-learn	Library for Machine Learning model development and evaluation
XGBoost	High-performance boosting algorithm for classification tasks
TensorFlow / Keras	Frameworks used for Deep Learning model development
SHAP	Explainable AI tool for interpreting model predictions
LIME	Provides local explanation for AI model decisions
Git	Version control system for tracking code changes
GitHub	Platform for team collaboration, project management, and code hosting